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Research Article

Dynamical Resilience Indicators in Time Series of Self-Rated Health Correspond to Frailty Levels in Older Adults

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Abstract

Background: We currently still lack valid methods to dynamically measure resilience for stressors before the appearance of adverse health outcomes that hamper well-being. Quantifying an older adult's resilience in an early stage would aid complex decision-making in health care. Translating complex dynamical systems theory to humans, we hypothesized that three dynamical indicators of resilience (variance, temporal autocorrelation, and cross-correlation) in time series of self-rated physical, mental, and social health were associated with frailty levels in older adults.

Methods: We monitored self-rated physical, mental, and social health during 100 days using daily visual analogue scale questions in 22 institutionalized older adults (mean age 84.0, *SD*: 5.9 years). Frailty was determined by the Survey of Health, Ageing and Retirement in Europe (SHARE) frailty index. The resilience indicators (variance, temporal autocorrelation, and cross-correlation) were calculated using multilevel models.

Results: The self-rated health time series of frail elderly exhibited significantly elevated variance in the physical, mental, and social domain, as well as significantly stronger cross-correlations between all three domains, as compared to the nonfrail group (all $P < 0.001$). Temporal autocorrelation was not significantly associated with frailty.

Conclusions: We found supporting evidence for two out of three hypothesized resilience indicators to be related to frailty levels in older adults. By mirroring the dynamical resilience indicators to a frailty index, we delivered a first empirical base to validate and quantify the construct of systemic resilience in older adults in a dynamic way.

Keywords: complex dynamical system—adaptive capacity—critical transitions—continuous monitoring—well-being

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Introduction

In the face of an aging population, the focus is shifting from treating disease to maintaining the functional ability that enables well-being at old age (1). Important to the maintenance of functional ability is an individual's capacity to adapt to physical, mental, and social challenges that inevitably occur during the course of life (2–4). Quantifying an aging individual's functional reserves can be approached from different angles, such as allostatic load, frailty, or multimorbidity (5–8). Measurements of each of these constructs have in common that they aim to statically measure (bio)markers under basal conditions to judge whether their levels are in the "normal" or optimal range. Whereas this static approach to assess reserve capacities or cumulative damage has merit, it is recognized that an individual's capacity to deal with challenges is fundamentally a dynamical phenomenon (9). Although consensus regarding the concept is still lacking in the field of medicine, resilience—defined for working purposes as one's capacity to restore optimal functioning following health stressors—is emerging to refer to this dynamic capacity of the complex human system (10).

In the context of medicine, a complex system can be an organ system, a physiological system or a human being as a whole (a "system

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of systems"). It has been shown to some extent that stimulus-response patterns are indicators of systemic resilience: one perturbs the system and evaluates the response, for example by measuring recovery times (11). Yet, this requires the application of dedicated tests, which may not be feasible or raise ethical issues. In addition, stimulus-response tests can only be applied at discrete points in time and do not consider continuous fluctuations of system parameters over time. The translation of complex dynamical systems theory to human functioning may offer a complementary approach to quantifying the resilience of human systems that circumvents some of these disadvantages.

Complex dynamical systems theory hypothesizes that any complex system, like a human being, is permanently subject to natural perturbations from the environment. When one continuously monitors system parameters, the system's dynamic responses can be captured. Although most natural perturbations may be small, zooming in on the "microdynamics" of system parameters may still give a good impression of the system's capacity to recover. Systems with low resilience display three characteristic changes in the measured dynamics. First, a lack of resilience leads to an increase in the variance of fluctuations of a system parameter measured over time (12). Second, the temporal autocorrelation of a parameter time series typically increases as states become more correlated with states on subsequent moments (12). Third, in a system that consists of multiple subsystems, different subsystems are hypothesized to become more mutually dependent as they lose resilience. Hence, deviations of parameters in different domains of the system will become more correlated and cross-correlations between different time series rise (13,14). To give an example at the level of the whole person, this may mean that the physical health of a person with low resilience will fluctuate more heavily (increased variance) and will recover more slowly from a dip (increased temporal autocorrelation) and that this person is more likely to feel also mentally and socially less well at the time of experiencing a physical dip (increased cross-correlation) as compared to a person with high resilience (Figure 1).

For several complex systems, these three indicators of declining resilience have been shown to be related to critical transitions to ecosystem crises (15) and abrupt climate change (16). Preliminary indications for their validity in humans have been found in mood dynamics, where these resilience indicators turned out to be valuable in assessing the likelihood of onset of depression (17,18). To validate them as potential measures of the resilience of aging humans, we evaluated the association of baseline frailty levels in older adults with the three hypothesized resilience indicators in time series of self-rated physical, mental, and social health monitored over a period of 100 days.

Methods

Participants and Setting

Self-rated health was monitored prospectively in 28 older adults aged more than 70 years living in four residential care facilities in Italy. The setting was chosen to achieve the highest possible adherence to the study protocol, since residential care facility staff members could remind and motivate the participants to keep participating daily during the study period of 100 days. To guarantee reliable and feasible data collection, participants were only included if cognitive impairment (Mini-Mental State Examination [MMSE] score > 25) and severe functional impairment (Katz's Activities of Daily Living [ADL] score ≥ 2) were absent. Simultaneous participation in another study was also an exclusion criterion. During the study period, six participants dropped out for practical reasons (e.g., moving to another residential care facility, being unable to fill out the daily questionnaires and long-term hospitalization), resulting in 22 participants who completed the entire research period of 100 days. This study was a secondary analysis of data from a study published previously (19).

Measurement of General Health and Frailty

Baseline physical and mental health status were constructed from the Short Form 12 (SF-12) health survey, one of the most commonly used instruments to assess general health status in the aged population (20), composed of six physical and six mental health items. Baseline social health was determined by a combination of the 6-item Lubben Social Network Scale (LSNS-6) that assesses social networks of older adults (21) and the 6-item Friendship Scale (FS) that determines the quality of social connections (22). All survey questions concerned an individual's health during the past 4 weeks.

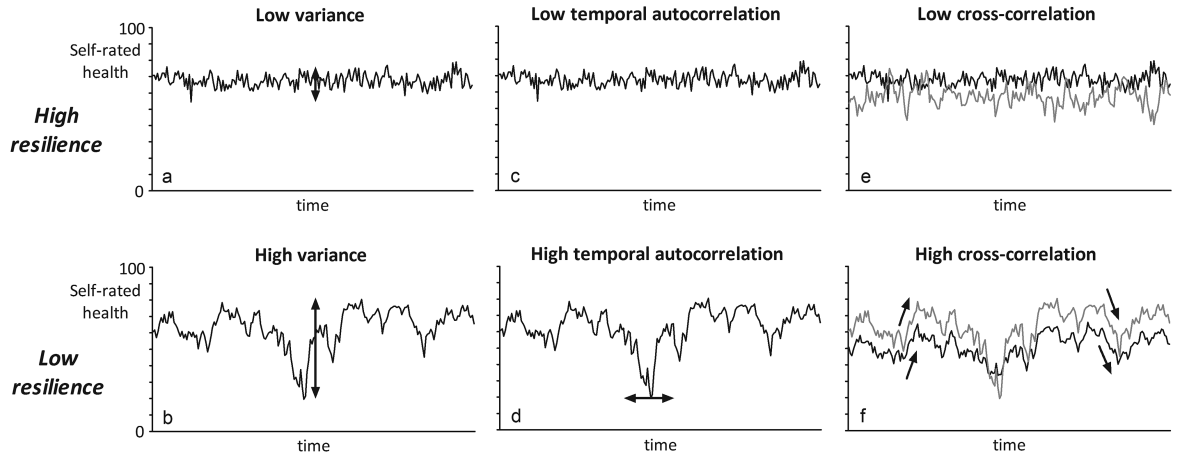

Figure 1. Graphical illustration of the dynamical resilience indicators. If a person has low systemic resilience, fluctuations of self-rated health over time are hypothesized to show higher variance (**a** vs **b**), higher temporal autocorrelation (**c** vs **d**), and higher cross-correlation between self-rated health domains (**e** vs **f**). In this example of fluctuations of self-rated health (data were randomly induced with a model), the black line may represent physical health and the gray line mental health.

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As a measure of frailty to mirror the approach of measuring resilience in this study, we used the Survey of Health, Ageing and Retirement in Europe (SHARE) frailty index (23). The SHARE frailty index has construct and predictive validity in comparable study populations and is freely accessible via web calculators. It defines frailty by five measures: exhaustion, loss of appetite, grip strength, walking difficulties, and physical activity. The SHARE frailty index provides a continuous score with higher scores corresponding to a frailer condition. In addition, it provides an associated categorical label (nonfrail, prefrail, and frail).

Measurement of Resilience Through Self-Rated Health Monitoring

The time series of self-rated health were acquired by using daily self-report questionnaires during 100 consecutive days. Every evening, participants had to answer six questions about their physical, mental, and social health during that day. Per domain, participants responded to an objective and a subjective question to acquire duplicate answers (see Supplementary Table 1). The objective questions evaluated the time spent during the day in specific conditions, while the subjective questions related to the perception of their health condition. Participants had to indicate their health ratings on 20 cm horizontal visual analogue scales. Responses could range from 0 (worst health possible) to 100 (best health possible). The questionnaires were provided in a structured diary that was previously tested in an empirical study (24). The data collection resulted in six different time series of 100 measurements in 22 participants. Of all 2,200 time points, only six time points were missing in one participant because of admission to the hospital.

Data Analyses

Based on the SHARE frailty index categorical labels, we divided our group of participants in two groups by combining those with the labels 'nonfrail' and 'prefrail' together to represent the nonfrail group (n = 11), which we could then compare with the frail group (n = 11). The SF-12, LSNS-6, and the FS scores were rescaled to a value between 0 and 1 with higher scores indicating better health status. Differences in baseline characteristics between nonfrail and frail participants were investigated in Stata version 11 using Wilcoxon rank–sum tests for continuous variables and Fisher's exact tests for categorical variables.

For the analysis of the resilience indicators, the data of self-rated health (SRH) scores collected with the visual analogue scales were first rescaled to a value between 0 and 1. At each time point, the ratings of objective and subjective health were averaged per domain (physical, mental, and social health). Subsequently, all data points within one time series were person-mean centered to exclude between-person mean SRH score differences. After that, three time series per participant remained (physical, mental, and social SRH during 100 days). Since the data have a hierarchical structure with repeated measures clustered within persons, multilevel regression models were used to calculate temporal autocorrelation (with model 1) and variance and cross-correlation (with model 2). We followed the model parameterization reported in an earlier paper that focused on the same type of research question (17). In the Supplementary Information, we elaborate on how we built the two different models in Stata version 11 and we included the syntax in Stata, SAS, and R.

Ethics

The local ethics committee of the University of Torino (Italy) granted permission and all participants provided informed consent.

Results

Description of Study Population

The analyses were performed using data from the 22 participants who completed the entire research period of 100 days. Table 1 shows their baseline characteristics. Fifty percent of the sample was identified as frail according to the SHARE frailty index, with a uniform distribution of frailty scores ranging from −0.7 (least frail) up to 4.7 (most frail) (Supplementary Figure 1). Frail elders had a significantly higher number of diseases and lower functional status, and had slightly lower cognitive functioning. The reported diseases were primarily related to physical functions; most frequently reported were hypertension (n = 11), osteoarticular disorders (n = 9), cardiovascular diseases (n = 9), respiratory diseases (n = 5), and neurological diseases (n = 5). Frail older adults scored significantly lower on the physical health survey than nonfrail participants, whereas baseline mental and social health status as determined by health surveys were comparable between the two groups.

Main Results: The Correlation Between Frailty and Resilience Indicators

The variance in the SRH score time series was significantly higher in frail participants across all three domains (Figure 2a): the variance in physical SRH time series was 0.0055 in nonfrail versus 0.0210

Table 1. Baseline Characteristics of the Sample

Characteristics	Total sample (n = 22)	Nonfrail (n = 11)	Frail (n = 11)	P value
Age in years, mean (*SD*)	84.0 (5.9)	82.5 (5.6)	85.5 (6.1)	0.34
Female, n (%)	15 (68%)	5 (45%)	10 (91%)	0.06
Low level of education (as opposed to high level), n (%)	10 (45%)	5 (45%)	5 (45%)	1.00
Number of diseases, mean (*SD*)	2.5 (1.3)	1.9 (0.7)	3.2 (1.5)	0.04
Cognition (MMSE), mean (*SD*)	28.2 (1.9)	29.0 (1.2)	27.5 (2.1)	0.05
Functional status (ADL), mean (*SD*)	4.5 (1.1)	5.3 (0.5)	3.8 (1.2)	0.001
Health status assessed by health surveys, mean (*SD*)				
Physical health	0.47 (0.19)	0.56 (0.17)	0.39 (0.18)	0.04
Mental health	0.51 (0.17)	0.52 (0.18)	0.51 (0.17)	0.95
Social health	0.55 (0.16)	0.55 (0.16)	0.56 (0.16)	0.74
Frailty score (SHARE frailty index), mean (*SD*)	1.9 (1.7)	0.5 (0.9)	3.4 (0.9)	<0.001

Note: ADL = activities of daily living (Katz) score (range: 0–6, lower is worse), MMSE = Mini-Mental State Examination (range: 0–30, lower is worse), SHARE = the Survey of Health, Ageing, and Retirement in Europe (higher is frailer).

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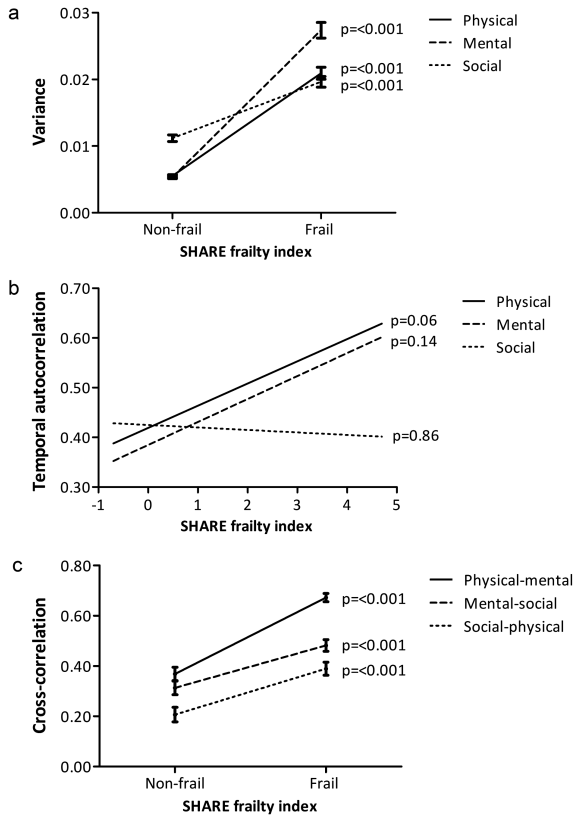

Figure 2. The resilience indicators of self-rated health time series (day 1–day 100) for nonfrail versus frail participants. (**a**) Variance in the physical, mental and social domain. (**b**) Temporal autocorrelation in the physical, mental, and social domain. (**c**) Cross-correlations between the physical-mental, mental-social, and social-physical domains. Error bars represent standard errors. For temporal autocorrelation, error bars are absent because the multilevel model that was used to analyze temporal autocorrelations did not allow for the estimation of standard errors. All exact values and *P* values can be found in Supplementary Table 2.

in frail participants ($P < 0.001$), and in mental SRH and social SRH 0.0053 versus 0.0274 ($P < 0.001$), and 0.0112 versus 0.0196 ($P < 0.001$) for nonfrail and frail participants, respectively.

Temporal autocorrelation in SRH scores showed no significant increase in frail participants for all domains (Figure 2b): the temporal autocorrelation in physical SRH time series was 0.44 in nonfrail versus 0.57 in frail participants ($P = 0.06$), and in mental SRH and social SRH 0.43 versus 0.54 ($P = 0.14$), and 0.42 versus 0.40 ($P = 0.86$), respectively for nonfrail and frail participants.

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All three cross-correlations between the health ratings of the physical, mental, and social domains were significantly stronger in frail participants (Figure 2c): the cross-correlation between physical-mental SRH time series was 0.37 in nonfrail versus 0.67 in frail participants ($P < 0.001$), and between mental-social SRH and social-physical SRH 0.31 versus 0.48 ($P < 0.001$), and 0.21 versus 0.39 ($P < 0.001$), respectively, for nonfrail and frail participants.

Discussion

Complex dynamical systems theory hypothesizes that three characteristics of time series (variance, temporal autocorrelation, and cross-correlations) serve as dynamical indicators of resilience. In this small sample of time series of self-rated health, we found two out of three resilience indicators to be related to frailty levels in older adults. Particularly variance in and cross-correlations between the time series of the physical, mental, and social domain were significantly increased in frail as compared to nonfrail older adults. Temporal autocorrelation was not significantly associated with frailty. Of the three measured health domains, the resilience indicators were most prominently elevated with frailty in the physical domain. These results provide first, preliminary evidence for the construct validity of these dynamical indicators as measures of resilience in aging individuals.

The fact that we did not find any significant results on temporal autocorrelation is in contrast with the previous study aimed on mood dynamics, which found a rise in temporal autocorrelation that was significantly associated with follow-up onset of depression (17). However, this effect was notably less prominent than for variance and cross-correlations. Since that study covered a much larger group of subjects, the current sample size of 22 participants may have been too small to result in significant differences in temporal autocorrelation. The fact that the strongest increase of dynamical resilience indicators was observed in the physical time series was expected, as our measure of frailty is based on five predominantly physical criteria. The SHARE frailty index is related to the frailty phenotype by Fried et al. (7), which captures mental factors to a lesser degree and does not cover social functioning (23).

The present paper responds to the World Health Organization's call for a shift from categorical measures of severe change in functional abilities to more sensitive measures that detect change early, as well as a shift from static thinking to monitoring individual trajectories over time (1). The resilience indicators applied in this study also address one of the key research topics in resilience research as formulated by Whitson et al. (2): measurement of physical resilience at the level of the whole person. Self-rated health is an established valid measure of individual health states in elderly (25–28) and we showed that self-rated health time series are suitable to use for an analysis of resilience indicators at the level of the whole person. This approach is in line with the emerging consensus that research on aging and frailty must focus on the dynamic interactions within and across systems (2,9,10). The potential value of translating the generic hypothesis of dynamical systems theory (12,29,30) to medicine has been pointed out before (31–33). With the current study, we deliver a first empirical validation of the construct of resilience indicators in human time series by relating them to frailty levels, in order to make a first step forward in the resilience research map laid out by Whitson et al. (2).

Besides the small sample size which may have caused this study to be underpowered for finding significant results on temporal autocorrelation, there is another notable limitation to this study. Our multilevel modeling method of analysis provides evidence based on between-subject variability: persons with elevated resilience indicators also belonged to the frail group. Critics have correctly argued that this type of analysis does not yet provide evidence for within-subject effects: a temporal association between rising resilience indicators and increasing frailty within single individuals (34,35). Whether the resilience indicators are sensitive enough to pick up changes in resilience within individuals and also have intraindividual predictive value for negative health outcomes remains to be studied. Such research questions require different study designs, such as the provocation study that found a rise in resilience indicators in mood time series within a single person before the onset of depression after

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stopping treatment with antidepressants (18). Future prospective exploration of resilience indicators in clinical settings should contribute to a better understanding of their validity and reliability (36).

For the clinical practice, this study opens a new door to dynamically measuring resilience for upcoming stressors that could complement the existing frailty measures. Potentially, when these resilience indicators would be evaluated not only in self-reported information but also in multiple physiological parameters of the human body at the same time, they could ultimately be used to rank a person from resilient to frail and to detect individual changes in resilience over time. Objectively quantifying an individual's resilience would allow more personalized and timely assessment of risk for critical transitions to health crisis states that often accompany old age, such as acute exacerbations of chronic diseases (e.g., acute heart failure decompensation, chronic obstructive pulmonary disease exacerbation) or geriatric syndromes (e.g., acute functional decline, delirium) (31–33,37,38). For example, a clinician may be more careful and take necessary precautions when considering to perform joint replacement surgery on an apparently vital older person with knee arthrosis knowing that this person has lost resilience. Identifying such seniors at highest risk of declining function was also recently termed a priority for emergency departments (39).

In conclusion, the current work is a first step toward quantifying systemic resilience in older adults in a dynamic way. Although preliminary, the current empirical validation of the construct against frailty offers an exciting prospect for further exploration of dynamical indicators of resilience in time series from humans. Eventually, resilience indicators may aid complex decision-making in health care of older adults and provide means to explore new opportunities for building and maintaining resilience.

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Supplementary Material

Supplementary data is available at *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences* online.

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Conflict of Interest

The authors have no conflict of interest to report.

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Dynamical Resilience Indicators in Time Series of Self-Rated Health Correspond to Frailty Levels in Older Adults

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